

Software Test Plan (STP) – Collaborative Study Material and Quiz Platform

Project: Collaborative Study Material and Quiz Platform

Version: 1.0

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1. Introduction

Purpose: This document defines the test plan for the Collaborative Study Material and Quiz Platform v1.0. It outlines the objectives, scope, testing strategy, resources, schedule, responsibilities, and criteria for testing the platform.

Scope: Testing covers user authentication and role management, study material sharing, quiz management, question bank management, performance analysis, collaboration and notifications, and the associated performance, reliability, usability, scalability, maintainability, and security requirements. Payment/subscription handling, plagiarism-detection engines, integration with external LMS platforms, and native mobile applications are excluded, as these are out of scope for this version of the platform.

References: Collaborative Study Material and Quiz Platform SRS v1.0; Software Architecture & Design Specification v1.0 (in progress).

Definitions: SRS (Software Requirements Specification), STP (Software Test Plan), FR (Functional Requirement), NFR (Non-Functional Requirement), RTM (Requirements Traceability Matrix), RBAC (Role-Based Access Control), JWT (JSON Web Token), MCQ (Multiple Choice Question), API (Application Programming Interface), UI (User Interface), CI/CD (Continuous Integration / Continuous Deployment).

2. Test Items

- User Authentication & Role Management module
- Study Material Sharing module
- Quiz Management module
- Question Bank module
- Performance Analysis module (Python analytics microservice)
- Collaboration & Notifications module
- Security features
- Performance, reliability, usability, scalability, and maintainability features

3. Features to be Tested

Functional requirements

- **CSQP-F-001:** User Registration
- **CSQP-F-002:** User Login (JWT issuance)
- **CSQP-F-003:** Role-Based Access Control
- **CSQP-F-004:** Password Reset
- **CSQP-F-010:** Upload Study Material
- **CSQP-F-011:** Browse & Search Material
- **CSQP-F-012:** View / Download Material
- **CSQP-F-013:** Bookmark Material
- **CSQP-F-014:** Edit / Delete Material
- **CSQP-F-020:** Create Quiz
- **CSQP-F-021:** Schedule Quiz
- **CSQP-F-022:** Attempt Quiz (Timed, Auto-submit)

- **CSQP-F-023:** Automatic Grading
- **CSQP-F-024:** Manual Grading of Subjective Answers
- **CSQP-F-025:** View Quiz Result
- **CSQP-F-026:** Export Quiz Results (CSV)
- **CSQP-F-030:** Add / Edit / Delete Question
- **CSQP-F-031:** Search & Filter Question Bank
- **CSQP-F-032:** Bulk Import Questions (CSV/Excel)
- **CSQP-F-033:** Duplicate Question Detection
- **CSQP-F-040:** Individual Performance Report
- **CSQP-F-041:** Class-level Analytics
- **CSQP-F-042:** Analytics Dashboard Charts
- **CSQP-F-043:** Export Performance Reports (PDF/CSV)
- **CSQP-F-050:** Comment on Study Material
- **CSQP-F-051:** In-app / Email Notifications
- **CSQP-F-052:** Rate Study Material

Non-functional requirements

- **CSQP-NF-001:** Response Time ≤ 2 seconds (90th percentile, 100 concurrent users)
- **CSQP-NF-002:** Scalability – 100 Concurrent Users without Degradation
- **CSQP-NF-003:** Reliability / Availability $\geq 99\%$
- **CSQP-NF-004:** Usability / Accessibility (WCAG 2.1 AA)
- **CSQP-NF-005:** Maintainability – 70% Unit Test Coverage

Security requirements

- **CSQP-SR-001:** Mandatory TLS 1.2+ Communication
- **CSQP-SR-002:** Secure Password Hashing
- **CSQP-SR-003:** JWT Access & Refresh Token Validation
- **CSQP-SR-004:** RBAC Middleware on Protected Routes
- **CSQP-SR-005:** Input Validation, Sanitization & Rate Limiting

4. Features Not to be Tested

- Payment / subscription handling
- Plagiarism-detection engines
- Integration with external LMS platforms
- Native mobile applications
- Internal workings of the third-party SMTP/email provider
- Internal workings of the cloud file storage provider (AWS S3 / Cloudinary / GridFS)

5. Test Approach / Strategy

Levels:

- **Unit Testing** — Testing individual modules such as authentication, material management, quiz management, question bank, and the Python analytics service.
- **Integration Testing** — Testing communication between the React frontend, Node.js/Express backend, MongoDB, and the Python analytics microservice.
- **System Testing** — Testing complete platform workflows from registration and login through material sharing, quiz attempts, and analytics.
- **Acceptance Testing (UAT)** — Verifying that the implemented system satisfies the acceptance criteria specified in the SRS.

Types:

- Functional Testing
- Regression Testing
- Performance Testing
- Usability Testing
- Security Testing
- Load / Scalability Testing

Entry Criteria: Stable build delivered, required test data available, test environment ready, database configured, and major modules available for testing.

Exit Criteria: All High-priority functional requirements are implemented and verified, no critical NFR or security failures remain, and the RTM shows all planned test cases passed.

5.1 Security Validation

- Verify TLS 1.2+ is enforced on all client-server and inter-service connections.
- Verify secure password hashing and ensure passwords are never stored or logged in plaintext.
- Verify expired or invalid JWT access tokens are rejected and the refresh flow issues a valid new token.
- Verify RBAC middleware rejects unauthorized-role requests on every protected route with HTTP 403.
- Test input validation and sanitization against injection/XSS attacks, and verify authentication endpoints are rate-limited.

6. Test Environment

Hardware:

- Standard computer / laptop
- Smartphone / tablet for responsive interface testing
- Internet connection

Software:

- React.js frontend
- Node.js with Express.js backend
- Python (Flask/FastAPI) analytics microservice
- MongoDB (local instance or MongoDB Atlas)
- Chrome

- Firefox
- Edge

Tools:

- VS Code
- Postman
- Browser Developer Tools
- MongoDB Compass
- JMeter / k6 (performance and load testing)
- OWASP ZAP (security scanning)

Test Data:

- Student accounts
- Instructor accounts
- Administrator account
- Sample study material files (PDF/DOCX/PPT)
- Sample quizzes and question bank entries
- Valid and invalid authentication credentials
- Valid and invalid input data

7. Test Schedule

Milestones:

- Test case design
- Environment setup
- Test execution start
- Test execution end
- UAT (aligned with the 28-09-2026 mini-project deliverable date)

8. Test Deliverables

- Test Plan
- Test Cases — Manual
- Test Data
- Test Execution Logs
- Defect Reports
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Sadanand	Prepare plan, coordinate execution
Test Engineer	Sagar K	Design & execute test cases, log defects
Developer	Sahana K S	Support defect fixes and triage
Product Owner	Sahana M S	Approve test results, sign-off readiness

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Request early smoke builds from the dev team
Test environment downtime	Maintain a backup environment on a cloud VM
MongoDB connectivity issues	Verify database connectivity before test execution and maintain backup test data
Python analytics service integration issues	Use a stubbed/mocked analytics endpoint for isolated frontend and backend testing
Authentication / security defects	Perform security testing before final acceptance
Insufficient test data	Prepare test accounts, material, quizzes, and questions before execution

11. Assumptions & Dependencies

- MongoDB (local instance or Atlas) will be available and accessible during testing.
- Required test data will be provided before test execution.
- The React frontend, Node.js/Express backend, and Python analytics service will be available for integration testing.
- Internet connectivity will be available during testing.
- Required authentication and authorization modules will be implemented before system testing.

12. Suspension & Resumption Criteria

Suspend testing if:

- Testing environment is unavailable for more than 4 hours.
- Build is too unstable and blocks more than 30% of planned test cases.
- Critical authentication or database failures prevent further testing.

Resume Testing if:

- Blocking defects are resolved.
- Testing environment is stabilized.
- A stable build is provided.

13. Test Case Management & Traceability

RTM ensures mapping of SRS requirements to test cases.

Example:

- * CSQP-F-002 (User Login) → TC-Auth-02
- * CSQP-F-003 (RBAC Enforcement) → TC-Auth-03
- * CSQP-F-010 (Upload Study Material) → TC-Mat-01
- * CSQP-F-020 (Create Quiz) → TC-Quiz-01
- * CSQP-F-022 (Attempt Quiz, Timed) → TC-Quiz-03
- * CSQP-F-030 (Manage Question Bank) → TC-QB-01
- * CSQP-F-040 (Individual Performance Report) → TC-Perf-01
- * CSQP-NF-001 (Response Time) → TC-Perf-NF-01
- * CSQP-SR-002 (Secure Password Storage) → TC-Sec-02

14. Test Metrics & Reporting

Metrics collected:

- % test cases executed
- % passed / failed
- Defect density
- Defect aging - Requirement coverage

Reports:

- Daily execution status
- Defect reports
- Final Test Summary Report

15. Detailed Test Cases

A representative set of test cases covering functional, non-functional, and security requirements, to be expanded during execution.

TC ID	Req Ref	Description / Steps	Expected Result	Priority	Type
TC-Auth-01	CSQP-F-001	Register a new user with valid name, email, password, and role (Student/Instructor).	Account is created and the user is redirected to the login page.	High	Functional
TC-Auth-02	CSQP-F-002	Log in with valid email and password.	A JWT and profile information are returned; user is authenticated.	High	Functional
TC-Auth-03	CSQP-F-003	As a Student, attempt to call an Instructor-only endpoint (e.g., create quiz).	Request is rejected with HTTP 403 Forbidden.	High	Functional
TC-Mat-01	CSQP-F-010	As an Instructor, upload a PDF tagged with course, subject, and topic.	File appears in the material library with the correct tags.	High	Functional
TC-Mat-02	CSQP-F-011	As a Student, search material by keyword.	Matching results are returned within 2 seconds for a seeded dataset.	High	Functional
TC-Quiz-01	CSQP-F-020	As an Instructor, create a quiz with MCQ, True/False, and short-answer questions, a duration, and a passing score.	Quiz is saved with all configured questions and settings.	High	Functional
TC-Quiz-03	CSQP-F-022	As a Student, attempt an active quiz and let the timer expire without submitting.	Quiz auto-submits at time-out and captures all answered questions.	High	Functional
TC-QB-01	CSQP-F-030	As an Instructor, add, edit, and delete a question with subject/topic/difficulty metadata.	Question is saved and retrievable by its metadata after each operation.	High	Functional
TC-Perf-01	CSQP-F-040	As a Student who has completed quizzes, open the individual performance report.	Report reflects all completed quiz attempts, grouped by topic.	High	Functional
TC-Perf-NF-01	CSQP-NF-001	Run a load test (e.g., JMeter/k6) with up to 100 concurrent users against page loads and APIs.	90th-percentile response time is $\leq$ 2 seconds.	High	Non-Functional
TC-Scale-01	CSQP-NF-002	Simulate 100 concurrent students attempting a scheduled quiz.	All attempts complete without service degradation or errors.	High	Non-Functional
TC-Sec-02	CSQP-SR-002	Register a user, then inspect the database for the stored password value.	Only a hashed value is present; plaintext password is never stored or logged.	High	Security

16. Approvals

Role	Name	Signature / Date
QA Lead	Sadanand	Sadanand 17/09/26
Dev Lead	Sahana K S	Sahana K S 17/09/26
Product Owner	Sahana M S	Sahana M S 17/09/26